FITFUEL
COMPLETE TEST REPORT

QA review
26 October 2026

Project: **my_first_app** | Flutter classroom demo for FitFuel calorie tracking

Repository layout: Flutter app at the repository root; design and QA material under docs/.

OVERALL RESULT	**4 widget tests passed | Static analysis passed | Web build passed**

Executive summary

The FitFuel source was reviewed and exercised with Flutter widget tests. The test suite covers a compact mobile viewport, empty credentials, invalid name and password boundaries, valid login, dashboard values, and logout. Static analysis reported no issues and the Flutter web build completed successfully. One app defect and two test-suite defects were reproduced and corrected.

The report distinguishes automated verification from manual browser verification. Flutter could not launch its managed Chrome instance in this environment, so the app was not visually inspected in a live browser. Native Android, iOS, Windows, macOS, and Linux builds were not run.

Execution summary

Check	Result	Evidence
Flutter analyze	**PASS**	No issues found; final run completed in 13.2 seconds.
Flutter widget tests	**PASS: 4/4**	All four tests passed after app and test corrections.
Flutter web build	**PASS**	Compiled successfully; web output created under build/web.
Live browser launch	**NOT VERIFIED**	Flutter Chrome launch failed after three attempts; no visual runtime claim is made.
Other platform builds	**NOT RUN**	Only the web target was built in this pass.

Environment

Flutter 3.47.7 (stable), Dart 3.13.5, Windows x64. The project has no app dependencies beyond Flutter and Cupertino Icons. The test runner required a writable temporary directory and local loopback access; after those environment constraints were handled, tests ran successfully.

Test coverage and results

ID	Scenario	Expected result	Result
TC-01	Submit the login form with both fields empty	Required-field messages appear; navigation is blocked.	**PASS**
TC-02	Name contains one character	Minimum two-character validation appears.	**PASS**
TC-03	Name contains a digit	Numeric-name validation appears.	**PASS**
TC-04	Password is shorter than six characters	Minimum-length validation appears.	**PASS**
TC-05	Use name Alex and password secret1	Dashboard opens with greeting, macro card, and 750 kcal remaining.	**PASS**
TC-06	Choose Logout from the dashboard	Start page is restored.	**PASS**
TC-07	Render the Start page at 320 x 568	Start content remains reachable without a RenderFlex overflow.	**PASS**

Defects found and corrected

Finding	Impact and evidence	Correction and verification
FTF-001 Obsolete starter test	The original widget test expected a counter and add icon. Those controls do not exist in FitFuel, so the supplied test did not exercise this app and was expected to fail.	Replaced the counter smoke test with FitFuel flow tests. The final suite passes 4/4.
FTF-002 Logout tap was off-screen	The first version of the new flow test attempted to tap Logout below the test viewport. The tap missed and the test failed; the dashboard itself is scrollable.	Updated the test to scroll Logout into view before tapping. The final flow test passes.
FTF-003 Start page overflowed on compact screen	At 320 x 568, a widget test reproduced a RenderFlex overflow of 44 pixels at the bottom of the Start page.	Made the page scrollable and sized the hero area for available height. The compact viewport test now passes without a layout exception.

The compact-screen overflow was an app defect and has been fixed in lib/main.dart. The other two findings were test-coverage/test-interaction issues; all three corrections are verified by the passing 4/4 test suite.

Reviewed application behavior

Area	Review notes
Login	Name is trimmed and must be at least two characters with no digits. Password requires six or more characters. Password entry is obscured.
Dashboard	Displayed sample values are static: goal 2,200 kcal, eaten 1,450 kcal, burned 420 kcal, and remaining 750 kcal.
Navigation	Start opens Login; valid demo credentials open Dashboard; Logout clears the route stack and returns to Start.
Compact layout	Start-page content now scrolls when the available height is short, and its hero drawing height adapts for compact screens. Verified at 320 x 568.

Scope, limitations, and next steps

Scope performed

Reviewed lib/main.dart, pubspec.yaml, the widget test suite, web shell/manifest, and project structure. Ran flutter analyze, flutter test, and flutter build web. The sibling APP folder contains SVG screen designs and was treated as design reference material rather than a runnable application.

Verification limits

Limit	Effect on conclusion
Live browser session	Flutter could not launch its managed Chrome after three attempts. The web build and widget tests passed, but visual layout and browser-console behavior were not verified.
Platform matrix	Android, iOS, Windows desktop, macOS, and Linux targets were not individually built or launched. Passing the web build does not establish native-platform readiness.
Backend/persistence	The source explicitly identifies this as a classroom demo with no backend or real authentication. Login accepts any values that meet validation; dashboard metrics are fixed sample data. These are documented demo constraints, not defects in the tested flow.

Recommended follow-up

1. Open the web build in a normal browser and inspect the Start, Login, validation-error, Dashboard, and Logout screens at mobile and desktop widths.

2. Build and launch each platform that is part of the class deliverable (for example, Android emulator/device or Windows desktop).

3. If this project is intended to be a real tracker, replace the classroom login and fixed calorie/macronutrient values with the agreed authentication, data-entry, and persistence behavior.

Final QA disposition	**The compact-screen overflow is fixed. Four widget tests, static analysis, and web compilation pass. Live visual behavior and non-web platform readiness remain unverified.**

Test source updated: my_first_app/test/widget_test.dart. Web build output: my_first_app/build/web. This report was prepared from the stated checks and source review; it does not certify untested devices or services.